Modeling Pragmatic Reasoning behind Social Meaning

[Anonymous for review]

Abstract

Researchers are increasingly interested in how pragmatic reasoning influences the social impressions we form of a speaker. This study examines the complex interplay between language use, situational context, pragmatic inference, and social meaning. The Social Evaluation Model (SEM) is presented, a probabilistic framework that models how listeners form social judgments about speakers based on their linguistic choices and the situational context, by reasoning about the speaker's knowledge states and communicative motivations. We situate SEM within the broader tradition of probabilistic pragmatics, showing how it extends the inferential chain of the Rational Speech Act framework to social evaluation as an explicit target of inference. We test the model's predictions against data from a matched-guise experiment on numerical (im)precision, evaluating both its qualitative and quantitative fit to human social judgments. Our findings contribute to existing research in sociolinguistics and formal pragmatics, providing a principled probabilistic framework for understanding how linguistic choices convey social meaning through pragmatic inference.

Keywords: social meaning, pragmatic inference, social evaluation, probabilistic pragmatics, Rational Speech Act, numerical precision

1 Introduction

1.1 From Gricean Maxims to Probabilistic Pragmatics

The study of pragmatic inference, how listeners derive meaning beyond the literal words, began with H. P. Grice's influential work on implicature (Grice, 1975). Grice proposed that conversation follows a *Cooperative Principle* along with maxims that speakers typically adhere to for effective communication. When a speaker seemingly violates a maxim, the listener assumes cooperation and infers an additional layer of meaning, known as *implicature*. This framework established that meaning relies on *rational inference* regarding the speaker's intent. Building on Grice's foundation, Horn (1989) and Levinson (2000) introduced *neo-Gricean* models, refining the types of implicature and incorporating scalar reasoning (e.g., "some" suggesting "not all"). These early theories demonstrated that pragmatic inference involves not just language but also social reasoning, shaped by expectations of cooperation and relevance.

Gricean pragmatics initially focused on cooperative reasoning, but later developments began to view implicature as indicative of the speaker's epistemic state: what the speaker knows, believes, or is committed to. A central insight of this tradition is that deriving a strong ("not all") inference from a weaker form requires more than the assumption of cooperativity; it requires an additional *competence assumption*, namely that the speaker is informed about whether the stronger alternative holds (Geurts, 2010; Sauerland, 2004; Schulz and Rooij, 2006). Without this assumption, an underinformative utterance licenses only a weak *ignorance* inference (the speaker does not know that the stronger alternative holds); with it, the listener can strengthen this to the conclusion that the speaker knows it to be false. This move from primary to secondary implicature, the *epistemic step* (Breheny, Katsos, and Williams, 2006; Sauerland, 2004), makes explicit that pragmatic enrichment is driven by the listener's model of the speaker's knowledge, not by the utterance alone. The scalar case is the classic illustration (Chemla and Singh, 2014; Spector, 2007): whether "some" implies "not all" depends on whether the speaker is taken to know the stronger alternative to be false or merely fails to affirm it. More broadly, this represented a shift from truth-conditional to epistemic interpretations of pragmatic reasoning, in which the listener's inference is fundamentally an inference about the speaker's mind.

Probabilistic and Bayesian models, including the Rational Speech Act (RSA) framework (Frank and Goodman, 2012; Goodman and Stuhlmüller, 2013; Potts et al., 2016), advanced this perspective by conceptualizing pragmatic inference as Bayesian reasoning over the speaker's epistemic beliefs and communicative goals. From this viewpoint, pragmatic inference is nuanced and probabilistic, reflecting uncertainty about the speaker's knowledge and intentions.

1.2 Intentions, Motives, and Communicative Goals

Beyond epistemic reasoning, a substantial body of work locates pragmatic inference in the recovery of speaker intentions and motives: the *why* behind an utterance. The foundations are philosophical. Grice (1957) defined speaker meaning itself in terms of intention recognition: a speaker means something by an utterance in virtue of intending the hearer to recognize that very intention, so that understanding an utterance is, at bottom, an inference about the speaker's communicative intentions. Sperber and Wilson's (1995) Relevance Theory develops this into a cognitive account on which comprehension is the search for an interpretation that the speaker could rationally have intended to be optimally relevant, recovering both the speaker's informative intention (what they aim to convey) and their communicative intention (that they aim to convey it). On these views, inference about meaning is inseparable from inference about why the speaker spoke as they did.

Crucially, the relevant motives are often *social* rather than purely informational. Brown and Levinson's (1987) politeness theory casts utterance choice as the management of face, and work in developmental and cognitive pragmatics treats communication as fundamentally cooperative social action (Clark, 1996; Tomasello, 2003). The clearest illustration that speakers choose forms strategically comes from research on indirect speech: Lee and Pinker (2010) explain off-record indirectness through a "strategic speaker" who trades efficiency for plausible deniability when unsure whether the hearer is cooperative or antagonistic, on the premise that communication mixes cooperation with conflict and that language serves both to convey information and to negotiate the speaker–hearer relationship. What such work establishes, and what matters for the model developed below, is that the choice of one form over an available alternative is itself an object of inference: listeners routinely ask not only what a speaker meant but why they chose to mean it *that way*, and they draw conclusions about the speaker accordingly.

Recent computational and cognitive models bring the epistemic and intentional strands together, treating pragmatic interpretation as joint Bayesian reasoning about what the speaker knows and what they are trying to do. The Rational Speech Act framework (Frank and Goodman, 2012; Goodman and Frank, 2016) models the listener as inverting a model of a goal-directed speaker, and Bergen, Levy, and Goodman (2016) show how reasoning over semantic alternatives and speaker goals can be combined within a single probabilistic system. Extensions to explicitly social goals, such as Yoon et al.'s (2020) model of polite speech, demonstrate that listeners jointly infer informational and social intentions from a single utterance. This convergence reflects a broader trend in contemporary pragmatics toward integrating informational (epistemic) and intentional (motivational) reasoning within one social-cognitive process, precisely the integration the present model builds on.

1.3 The Role of Context

Both strands of pragmatic reasoning surveyed so far, the epistemic and the motivational, operate under the control of context. A large experimental literature shows that even the most-studied inferences are not computed automatically but depend on the situation: scalar implicatures arise more readily when the context makes the stronger alternative relevant, and recede when it does not (Breheny, Katsos, and Williams, 2006; Degen and Tanenhaus, 2015). A recurring way of articulating this dependence is through the *Question Under Discussion*: context fixes which alternatives are live, and hence which inferences a listener is licensed to draw (Roberts, 2012). More generally, Rooij (2003) and Clark (1996) stress that pragmatic reasoning unfolds within common ground and joint activity, so that the same utterance can support different inferences depending on the conversational situation. Context, on this view, is not a backdrop but a determinant of pragmatic meaning: it sets the alternatives, the relevant question, and the assumptions against which an utterance is interpreted.

This context-sensitivity is not peculiar to epistemic inference. As the next subsection develops, the *social* meaning a listener reads off a speaker's choice is equally situation-dependent: the same linguistic form can support different social interpretations in different contexts, rather than carrying a fixed social value (Campbell-Kibler, 2007). It is this shared property, that both what a speaker is taken to know and how they are socially evaluated depend on the context of utterance, that the model developed in this paper builds on, treating social evaluation as a context-sensitive inference of the same general kind that governs epistemic interpretation.

1.4 Social Meaning and Pragmatic Reasoning

Recent research in sociopragmatics explores how pragmatic reasoning interacts with *social meaning*: the information language provides about a speaker's social identity, affiliations, and personal traits (Eckert, 2008; Podesva, 2011; Silverstein, 2003). The study of social meaning has its roots in variationist sociolinguistics. Labov's (1972) foundational work demonstrated that linguistic variation is systematic and socially stratified, linking language use to social structure and change. Subsequent research moved beyond correlating variants with fixed demographic categories toward understanding how speakers actively deploy variation to construct stances, styles, and personas; Eckert's (2012) "three waves" synthesis characterizes this trajectory, culminating in a view on which social meaning is not a static property of a form but is produced in interaction.

On this view, a linguistic variant is not tied to a single fixed social value. Eckert (2008) and Silverstein (2003) argue that a variant indexes an *indexical field*, a constellation of related potential meanings, only some of which are activated on any given occasion, depending on the speaker, the listener, and the situation. Campbell-Kibler (2007) demonstrates this empirically: the English variable (ING) shifts listeners' perceptions of a speaker along dimensions such as education, articulateness, and formality, but its effect is not constant; the same variant can be heard as, for instance, compassionate or condescending depending on the listener's prior orientation to the speaker, and its social impact depends on the surrounding linguistic and situational context. This is the context-sensitivity anticipated at the end of the previous subsection: social meaning, like epistemic meaning, is the product of situated interpretation rather than a fixed mapping from form to value.

A terminological clarification is in order, since "social meaning" is used in two related senses in the literature this work draws on. The first is the *indexical* sense just described: the marking of a speaker's social identity, group membership, or persona through conventionalized associations between linguistic form and social category. A second sense, prominent in recent work in semantics and pragmatics, concerns the *attributes* a listener ascribes to a speaker, traits such as competence, warmth, or pedantry, but also transient stances and affective states, on the basis of a particular linguistic choice (Beltrama, 2020; Beltrama and Staum Casasanto, 2021). The present work is concerned with social meaning in this second, attribute-based sense: SEM models how a listener arrives at trait judgments of a speaker from a pragmatic choice, not how linguistic forms come to index social categories. The two senses are not separate phenomena so much as regions of a continuum: as Eckert (2019) argues, social meaning ranges from stable social categories to flexible, interaction-based stances, and Beltrama (2020) characterizes social meaning as a constellation of traits (demographic, personal, and ideological) that linguistic forms convey about their users. Attribute-level inferences of the kind SEM models are thus plausibly among the ingredients from which richer identity-level meanings are built; SEM isolates and formalizes this attribute-based step.

The attribute-based sense of social meaning has recently become the focus of a growing body of work at the interface of pragmatics and sociolinguistics, which shares a common premise: that many social meanings are not read off the form directly, as in the indexical case, but are *derived inferentially* from the speaker's pragmatic choices (Acton, 2019; Beltrama, 2020). On this view, the social attributes a listener ascribes to a speaker, such as competence or warmth, are the product of reasoning about why the speaker made a particular choice among alternatives, in much the same way that epistemic implicatures are. Several strands of work develop this idea. Acton (2019) shows that even a function word like the definite article carries social meaning through the stance it signals toward a referent. Closest to the present study, Beltrama, Solt, and Burnett (2022) find that the choice between precise and approximate numerical expressions leads listeners to distinct social evaluations, modulated by context, and Beltrama and Papafragou (2023) show that a speaker's adherence to or violation of the Gricean maxims of Relevance and Informativeness shifts perceptions of their warmth and competence. Critically, the latter find that the social effect depends on the inferred *reason* for the speaker's behavior, for instance whether an uninformative answer is attributed to inability or to unwillingness, demonstrating that listeners' social judgments are driven by inferences about the speaker's epistemic state and motivation, not by the utterance alone. This is precisely the inferential route from pragmatic choice to social evaluation that the present work sets out to formalize.

1.5 The Present Work

Taken together, these developments reveal that pragmatic understanding is a form of social cognition: to interpret an utterance, a listener reasons jointly about what the speaker knows, why they chose to speak as they did, and what this implies about them as a social agent, all under the control of context. What has so far been established largely through separate experimental and theoretical strands, however, has not been brought together in a single explicit model of how these inferences combine to yield a social evaluation.

Building on these insights, we introduce the Social Evaluation Model (SEM), a probabilistic framework that formalizes how linguistic choices, situational context, and pragmatic reasoning work together to shape listeners' social judgments. SEM shows how listeners infer a speaker's knowledge state and communicative motivation from their language use, and how these inferences give rise to evaluations of social attributes such as competence, helpfulness, and likability. The framework expands the inferential chain of probabilistic pragmatics to include evaluative meaning, where small differences in language use can lead to subtle, context-dependent changes in social interpretation (e.g., Beltrama and Papafragou, 2023).

The paper is organized as follows. Section 2 provides a formal definition of the SEM, situating it within the probabilistic pragmatics tradition and comparing it to the RSA framework. A case study on numerical (im)precision is used to develop and qualitatively evaluate the model in Section 3. Section 4 presents a quantitative evaluation of the model's predictions, including a comparison of the full model against nested ablations. Section 5 discusses the model's contributions and directions for future work.

2 The Social Evaluation Model

2.1 From Meaning Recovery to Social Evaluation

The Social Evaluation Model (SEM) is a probabilistic framework for predicting how listeners form social judgments about speakers based on their linguistic choices. It builds on the core architecture of probabilistic pragmatics, in particular the Rational Speech Act framework (Frank and Goodman, 2012; Goodman and Frank, 2016) and the broader tradition of Bayesian pragmatic modeling surveyed by Franke and Jäger (2016), while extending that architecture toward a target of inference that has not previously been formalized within it: the social evaluation of the speaker as a communicative agent.

Like RSA and related models, SEM treats pragmatic interpretation as Bayesian inference. A listener who hears utterance u in context c does not simply decode its semantic content; they reason probabilistically about what mental states could have produced that utterance. Central to this reasoning is a set of utterance alternatives U, the other things the speaker could have said, and a commitment to the truthfulness constraint, which limits the knowledge states a speaker could plausibly be in given what they actually said. These features are not incidental borrowings from RSA; they are load-bearing components of SEM, and the model's predictions depend on them in the same structural way RSA predictions do.

Where SEM departs from existing probabilistic pragmatic models is in what the listener is ultimately trying to infer. In RSA, inference terminates at meaning: the listener recovers the speaker's intended message. In SEM, meaning recovery is not the endpoint but an intermediate step. The listener uses their inferences about the speaker's epistemic state, what the speaker knew and what they could therefore truthfully have said, to make a further inference about the speaker's communicative motivation: why did they choose this utterance over its alternatives? And from that motivational inference, together with the inferred epistemic state itself, the listener arrives at a social evaluation: what does this choice reveal about the speaker's competence, helpfulness, or friendliness?

This reasoning, inferring the knowledge state from the utterance, then the motivation from the knowledge state and context, is what SEM formalizes; the social attribute is then evaluated from both the inferred knowledge state and the inferred motivation jointly. The key novel components, relative to RSA, are: (i) social attributes as an explicit target of inference; (ii) discrete motivation-based strategies, each defined as a function from context and knowledge state to utterance, which are more enumerable and empirically grounded than the utility functions in RSA; (iii) an impact function $I(a|k, m; \omega)$ that links epistemic and motivational inferences to graded social evaluations; and (iv) empirically fitted trivalent functions that specify the direction of each inference-to-evaluation link. There is no counterpart to this function in standard RSA, because RSA does not model social evaluation as an output. Figure 1 provides an overview of the shared and novel components, and Figure 2 shows the full model as a Bayesian network.

2.2 The Model Architecture

The social evaluation model[1] (SEM) predicts the listener's social assessment of the speaker based on using utterance u from a set of alternatives U, given a situational context $c \in C$. Formally, the listener's evaluation function $E_L(a|u, c)$,[2] which specifies how the listener evaluates a speaker concerning social attribute a, is defined as:

[1] A Python implementation of SEM, including a reviewer-facing Jupyter notebook that reproduces the computations and figures in this paper, is available at `[ANONYMIZEDGITHUBREPOLINK]`.

[2] The function is defined as $E_L : A \times U \times C \to \mathbb{R}$; it returns a numerical score representing the listener's evaluation of the speaker on attribute a.

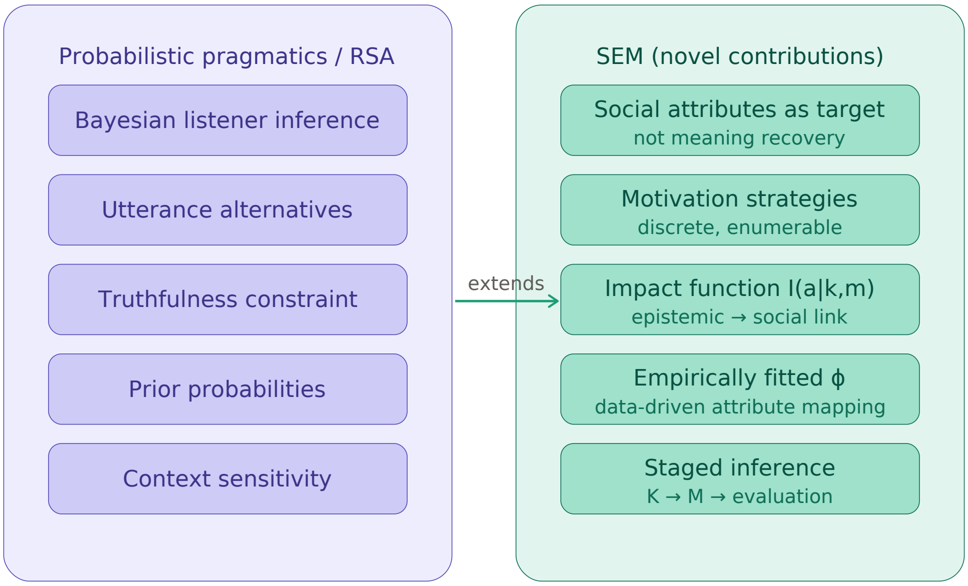


Figure 1: The relationship between Probabilistic Pragmatics / RSA and the Social Evaluation Model (SEM). SEM extends the core components of RSA (Bayesian listener inference, utterance alternatives, truthfulness constraint, prior probabilities, and context sensitivity), with novel components: social attributes as the target of inference, discrete motivation strategies, an impact function $I(a \mid k, m; \omega)$ linking epistemic to social dimensions, empirically fitted trivalent functions ϕ, and staged inference $(u, c \to K \to M \to a)$.

$$E_L(a|u,c) = \sum_{k \in K} \sum_{m \in M} P(k|u) \cdot P(m|k,u,c) \cdot I(a|k,m;\omega) \tag{1}$$

where K is a set of speaker knowledge states, M is a set of motivation-based strategies,[3] $P(k|u)$ and $P(m|k,u,c)$ are conditional probabilities, and $I(a|k,m;\omega)$ is an impact function that determines how inferences about knowledge state and motivation jointly affect the listener's evaluation of social attribute a. The weight factor ω adjusts the relative contribution of epistemic versus motivational inference to the evaluation.

This equation can be understood as formalizing two intuitive notions: (i) when a listener hears utterance u in context c, they infer the probability of the speaker being in knowledge state k and the probability of the speaker acting according to motivation strategy m; and (ii) the listener holds an internal mapping from knowledge states and

[3] A motivation-based strategy is a function that specifies what utterance a speaker would produce across different contexts and knowledge states when driven by a particular motivation. The same behavioral strategy can in principle arise from different underlying motivations; the case study in Section 3 provides detailed examples.

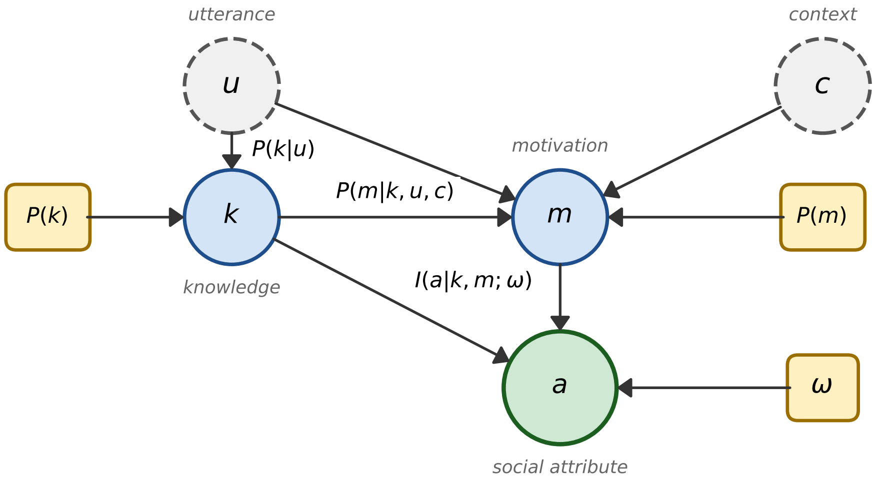


Figure 2: Bayesian network representation of the Social Evaluation Model. Dashed circles denote observed variables (utterance u, context c) available to the listener. Solid circles denote latent variables inferred by the listener: knowledge state k, motivation-based strategy m, and social attribute evaluation a. Rectangles denote fixed parameters: prior probabilities $P(k)$ and $P(m)$, and the weight parameter ω. Inference proceeds in three stages: u and $P(k)$ jointly determine the posterior over knowledge states $P(k|u)$; k, u, c, and $P(m)$ determine the posterior over strategies $P(m|k, u, c)$; and k and m feed into the impact function, weighted by ω, to yield the social evaluation a. The dashed arrow from k to a reflects the direct contribution of epistemic inference alongside the motivational pathway via m.

motivations to social attributes, captured by the impact function I. The evaluation function E_L returns a real number $x \in [-1, 1]$, where $x = -1$ indicates the lowest evaluation and $x = 1$ the highest.[4]

Knowledge states and the truthfulness constraint. A central aspect of SEM is the set U, which contains the utterances a speaker considers as alternatives in a given situation. SEM focuses specifically on *non-equivalent* alternatives, i.e., forms that differ in their core-semantic meaning. This is important because semantic differences between alternatives create distinct constraints on what knowledge state the speaker could plausibly be in: a speaker who uses a semantically weaker form (e.g., "some students passed") may or may not know the stronger fact, whereas a speaker who uses a semantically stronger

[4]When a categorical judgment is required (e.g., does the speaker possess attribute a or not), one applies the sign function: $x_{cat} = \text{sgn}(x)$. For a probabilistic output, one applies the sigmoid: $x_{prob} = (1+e^{-\kappa x})^{-1}$, where $\kappa > 0$ controls the steepness.

form (e.g., “some but not all students passed”) is committed to knowing it.

These different epistemic possibilities are represented by the set of knowledge states K. Inferences about knowledge states are formalized by the conditional probability $P(k|u)$, derived from the assumption that the speaker adheres to the maxim of truthfulness: they only assert what they know to be true. This assumption is implemented via a truthfulness function $t : K \to \mathcal{P}(U)$, which maps each knowledge state k to the set of utterances the speaker could truthfully produce in that state. The posterior probability of knowledge state k given utterance u is then:

$$P(k|u) = \frac{P(k) \cdot \chi_t(k,u)}{\sum_{k' \in K} P(k') \cdot \chi_t(k',u)} \tag{2}$$

where $P(k)$ is the prior probability of knowledge state k, and $\chi_t(k,u) = 1$ if $u \in t(k)$ and 0 otherwise.

Motivation-based strategies. The set M represents the motivation-based strategies a speaker might follow. Each strategy $m \in M$ is a function $m : C \times K \to U$ that specifies the utterance a speaker would choose in context c with knowledge state k when driven by a particular motivation. For example, a maximally informative strategy always selects the most precise available utterance given the current knowledge state, while a speaker-convenience strategy always selects the least effortful form regardless of context.

Given a set of strategies, the listener can calculate $P(m|k,c,u)$, the conditional probability that the speaker follows strategy m given knowledge state k, context c, and observed utterance u:

$$P(m|k,c,u) = \frac{P(m) \cdot \chi_m(c,k,u)}{\sum_{m' \in M} P(m') \cdot \chi_m(c,k,u)} \tag{3}$$

where $P(m)$ is the prior probability of strategy m, and $\chi_m(c,k,u) = 1$ if $m(c,k) = u$ and 0 otherwise.

The impact function. The impact function $I(a|k,m;\omega)$ defines how inferences about knowledge state k and motivation strategy m affect the social evaluation on attribute a.

It captures the intuition that listeners' assumptions about what a speaker knew and why they spoke as they did are systematically linked to evaluations of social attributes. For example, inferring that a speaker knew a precise value may positively affect perceptions of competence, while inferring that a speaker chose to omit relevant information may negatively affect perceptions of warmth. Formally:

$$I(a|k, m; \omega) = \omega \cdot \phi_K(k, a) + (1 - \omega) \cdot \phi_M(m, a) \tag{4}$$

where $\phi_K : K \times A \to \mathbb{T}_3$ and $\phi_M : M \times A \to \mathbb{T}_3$ are *trivalent functions* that indicate whether and how a knowledge state or motivation-based strategy influences a social attribute.[5] The weight parameter $\omega \in (0, 1)$ controls the relative contribution of epistemic versus motivational inference: high ω places more weight on what the speaker knew; low ω places more weight on their inferred motivation.

A note on whose reasoning these parameters encode. The free parameters of SEM, the priors $P(k)$ and $P(m)$ and the weight ω, are properties of the *listener*, not the speaker: they specify how a particular listener distributes prior expectation over knowledge states and motivations, and how heavily they weigh epistemic against motivational inference in arriving at a social judgment. Different listeners may bring different priors and weightings to the same utterance, and SEM represents this variation directly through these parameters. The model as stated thus describes the inferential process of a single, parameterized listener; a population of listeners corresponds to a distribution over parameter settings. We exploit exactly this interpretation in Section 4.3, where sampling over the parameter space functions as a virtual experiment across a range of listeners.

Relation to RSA. SEM is best understood as a distinct model that *extends* the RSA inferential chain rather than as a competitor to it. Frank and Goodman's (2012) framework could in principle be extended to include social attributes as additional latent variables, but doing so would still require the two components that constitute SEM's substantive

[5] A trivalent function maps inputs to one of three values in $\mathbb{T}_3 = \{-1, 0, 1\}$, representing a significant positive effect, no significant effect, or a significant negative effect respectively. These values are fitted empirically from participant data, as detailed in Section 3.

contribution: a set of discrete, empirically groundable motivation strategies in place of RSA's utility-based speaker model, and the impact function I with its trivalent mappings ϕ_K, ϕ_M, which have no counterpart in standard RSA. We defer a fuller comparison of the two architectures to Section 5, where we also discuss how SEM and RSA can be integrated, with SEM's social evaluations serving as inputs to a downstream speaker model.

3 A Case Study on Numerical (Im)Precision

We assess the Social Evaluation Model (SEM) by examining the phenomenon of numerical (im)precision, which refers to the level of detail or granularity at which numerical information is conveyed (Krifka, 2009). For example, a speaker who wishes to communicate the time at which an event occurred might use either of the alternatives shown in (1):

(1) a. "The accident happened at 8:32." (precise)

b. "The accident happened around 8:30." (approximate)

If a speaker has only approximate knowledge of the relevant facts, then if speaking truthfully they can only use the approximate alternative (1b); but a speaker who knows the precise value might express themselves either precisely or approximately, making this a situation of speaker choice.

Numerical (im)precision serves as an ideal testing ground for SEM, highlighting how linguistic choices reflect both contextual adaptation and social interpretation. It is also a domain where probabilistic pragmatics has already proven productive: Kao et al. (2014) model the nonliteral interpretation of number words, including the "pragmatic halo" by which round numbers are understood imprecisely, as Bayesian social inference about the speaker's communicative goals, within the RSA framework. Their model targets what an imprecise number is taken to *mean*; SEM addresses the complementary question of what the choice of an (im)precise form reveals about the *speaker*. Beyond interpretation, previous research has shown that speakers' preferred degree of precision varies with contextual demands and communicative relevance (Mühlenbernd and Solt, 2022; Van der Henst, Carles, and Sperber, 2002), and that different levels of (im)precision convey distinct social meanings (Beltrama, 2018; Beltrama, Solt, and Burnett, 2022); listeners' social evaluations are in turn linked to their inferences about a speaker's knowledge state and communicative motivations (Author, 2025). These findings suggest that (im)precision offers a rich domain for exploring how epistemic and contextual factors jointly shape pragmatic and social inference.

Van der Henst, Carles, and Sperber (2002) provided a foundational account of how context influences numerical precision, finding that speakers adjust their level of precision to match what is communicatively relevant: round numbers were preferred when exact values were pragmatically unimportant, while precise numbers were used when accuracy mattered. Mühlenbernd and Solt (2022) extended this in a production study where participants systematically adapted their expressions to context: more exact expressions when accuracy was pragmatically important, and more approximate ones when fine detail was unnecessary.

Another line of research reveals that numerical (im)precision conveys social meaning. Beltrama (2018) demonstrated that precise non-round expressions can signal positive attributes like intelligence as well as negative ones such as uptightness. Beltrama, Solt, and Burnett (2022) found that speakers using precise forms were rated higher on status-related attributes (e.g., 'intelligent', 'articulate') but also on 'pedantic', whereas approximate speakers received higher ratings on solidarity-related attributes (e.g., 'likeable', 'laid-back'). These evaluative patterns were context-dependent: the status-enhancing effect of precise forms was stronger in formal "for-the-record" contexts than in casual "bonding" contexts. These findings highlight exactly the kind of interplay between pragmatic expectations, social goals, and contextual norms that SEM aims to capture.

We will use SEM to predict the empirical results of an experiment we conducted, reported in full and preregistered in Author (2025) (preregistration: `[ANONYMIZEDOSFPREREGLINK]`). This experiment investigated how situational context influences the social meaning of precise and imprecise numerical forms. We employed the matched guise methodology (Campbell-Kibler, 2007), where participants evaluate speakers in different versions or ‘guises’ of the same scenario. In what follows, we summarize the experimental design and results relevant to fitting and testing SEM, then present the model predictions.

3.1 Participants, Design, and Methods

A total of 371 native English speakers, aged 18–64, with U.S. IP addresses took part via Prolific (payment: £1.20). The experiment used six brief, pre-tested scenarios in which one speaker asks a question and another responds with a numerical answer. Two factors were manipulated in a 2×2 design: the required precision level (HPR: high, LPR: low) and the numerical form (PRECISE or APPROX). An example scenario is shown in Figure 3. The study used a fully between-subjects design ($n \approx 90$ per condition), with each participant assigned one of the six scenarios in one of its four versions. Participants completed two tasks relevant to this study:

- **Task 1:** Participants rated the speaker on six social attributes using a 7-point Likert scale: COMPETENT, KNOWLEDGEABLE, WELL-PREPARED, LIKEABLE, HELPFUL, and PEDANTIC.
- **Task 2:** Participants indicated the speaker’s inferred reason for using a precise [imprecise] expression rather than a more approximate [more precise] alternative, via a free-text fill-in-the-blank format.

Author (2025) additionally report a second experiment that manipulates established speaker knowledge state; since the present model evaluation draws on the context/form data above, we do not make further use of it here and refer the reader to that paper for details.

3.2 Experimental Results

All analyses were performed using R (R Core Team, 2021), with linear mixed effects models fitted via *lmerTest* (Kuznetsova, Brockhoff, and Christensen, 2015), using form, context, and their interaction as fixed factors, random intercepts for scenario, and sum

LPR: Jamie has a new bicycle and is telling a friend about it. The friend is interested and wants to know more. *Friend:* "How much did the bicycle cost? I'd love to get one like it."	**HPR**: Jamie's new bicycle was stolen. Fortunately it was insured, so Jamie has called the insurance company. *Insurance agent:* "How much did the bicycle cost? I'll start the paperwork right away."
Jamie: "The bicycle cost **$509.55** (**PRECISE**) / **about $500** (**APPROX**)."	

Figure 3: Example scenario (LPR: low precision context; HPR: high precision context) with the numerical forms PRECISE and APPROX. This 'stolen bike' scenario is one of six pre-tested scenarios used in the experiment.

coding for predictors. The results of Task 1 for the attributes COMPETENT, LIKEABLE, and PEDANTIC are presented in Figure 4 (mean values and standard deviations).[6]

For COMPETENT and PEDANTIC, a significant main effect of form was found, with PRECISE rated higher ($p < .001$ for both). For COMPETENT and LIKEABLE, a significant interaction of context and form was found: the relative strength of PRECISE versus APPROX was greater in HPR than in LPR ($p < .001$ for both). These results demonstrate that the choice of numerical precision level carries social meaning: precise speakers score higher on competence-related dimensions compared to approximate ones, though they are also perceived as more pedantic. These effects are context-dependent: the positive attributes associated with precise forms are heightened in high-precision situations, while positive associations with speaking approximately are more pronounced in low-precision contexts.

Figure 5 displays the results of Task 2, showing frequencies of the 10 most common motivation classes, color-coded by form (blue: approximate, red: precise). Two coders independently coded the free-text responses; a brief description of each motivation class is provided in Figure 7(a).

To explore whether inferred speaker motivations influence social meaning, a core assumption of SEM, we compared, for each Task 2 motivation class, the Task 1 ratings of participants who did versus did not mention that class, using independent-samples t-tests

[6] The attributes KNOWLEDGEABLE, WELL-PREPARED, and HELPFUL showed closely aligned statistics with COMPETENT across all tasks (see Author, 2025, for full results). We therefore exclude them from individual treatment in the SEM evaluation, treating them as encompassed by COMPETENT.

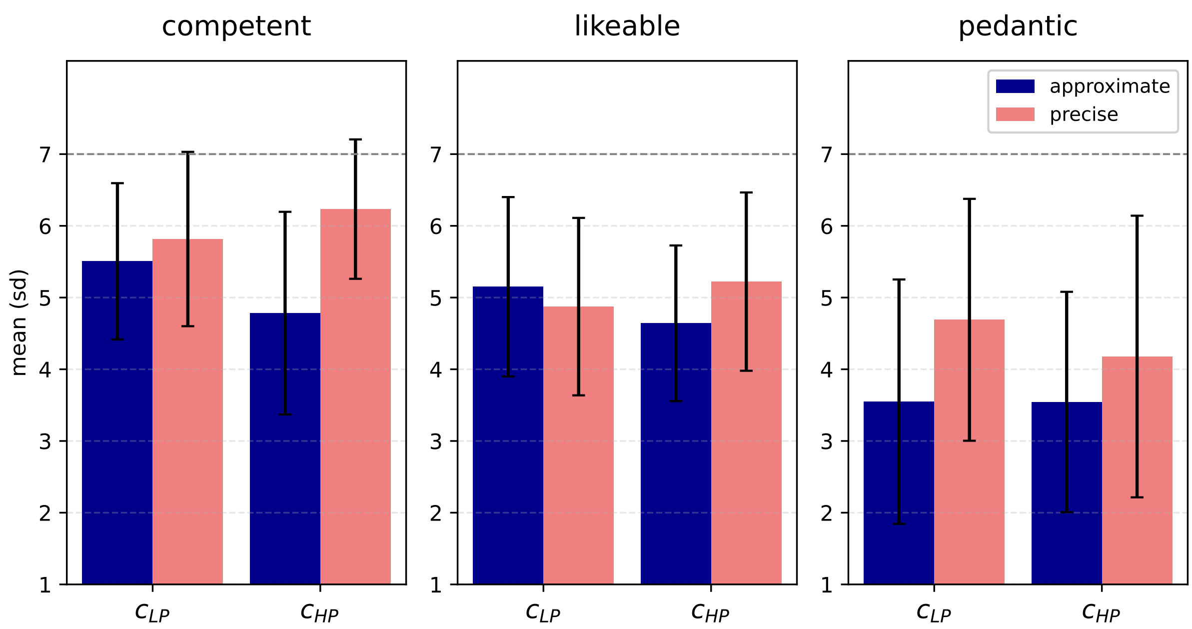


Figure 4: Task 1 rating results (Experiment 1): mean ratings with standard deviations for the attributes COMPETENT, LIKEABLE, and PEDANTIC, by numerical form and context. The dashed line marks the top of the 7-point rating scale.

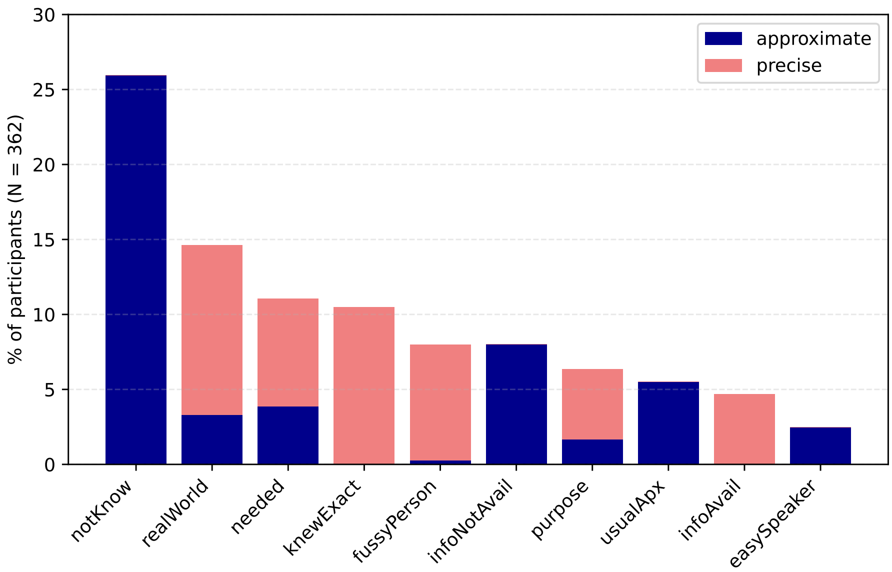


Figure 5: Relative frequencies (as a percentage of all participants) of the 10 most frequently mentioned motivation classes, color-coded by form: approximate (blue), precise (red).

(Experiment 1). Figure 7(b) displays the p-values for all combinations (3 social attributes $\times$ 10 motivation classes), with significant effects ($p < .01$) in bold font and direction indicated by color (green: positive, red: negative). A total of 7 significant effects were found across 30 combinations, supporting the central SEM assumption that social meaning is influenced by assumed speaker motivations.

3.3 Fitting the Model

To fit SEM to this case study, we construct a scenario model that matches the experimental design. We define two situational contexts: c_{LP} (low-precision needs) and c_{HP} (high-precision needs), and two utterance types: u_{prc} (precise) and u_{apx} (explicitly approximate). Figure 6 illustrates the model using an analogous scenario in which a speaker reports the time of an event either precisely ("at 8:32") or approximately ("around 8:30"), to either a friend at a dinner party or a police officer.

We consider two relevant knowledge states: k_p, the speaker has precise knowledge (e.g., knows the event happened at 8:32), and $k_{\neg p}$, the speaker has only approximate knowledge (e.g., knows it happened around 8:30 $\pm$ 5 min). The truthfulness function t is defined as $t(k_p) = \{u_{apx}, u_{prc}\}$ and $t(k_{\neg p}) = \{u_{apx}\}$: a speaker with only approximate knowledge cannot truthfully use a precise utterance. This is shown in Figure 6(a).

We consider three motivation-based strategies, shown in Figure 6(b):

- m_{max}: the speaker is always maximally precise, as long as their knowledge state permits it (always precise if k_p, approximate if $k_{\neg p}$).
- m_{sit}: the speaker matches their precision to the situational demands (precise in c_{HP}, approximate in c_{LP}, as long as knowledge permits).
- m_{min}: the speaker is always minimally precise regardless of context or knowledge.

Figure 6(c) illustrates the listener's inferences after hearing a precise utterance: the listener concludes the speaker has precise knowledge (k_p), and in a high-precision context assigns positive probability to both m_{max} and m_{sit}, while in a low-precision context infers the speaker follows only m_{max}.

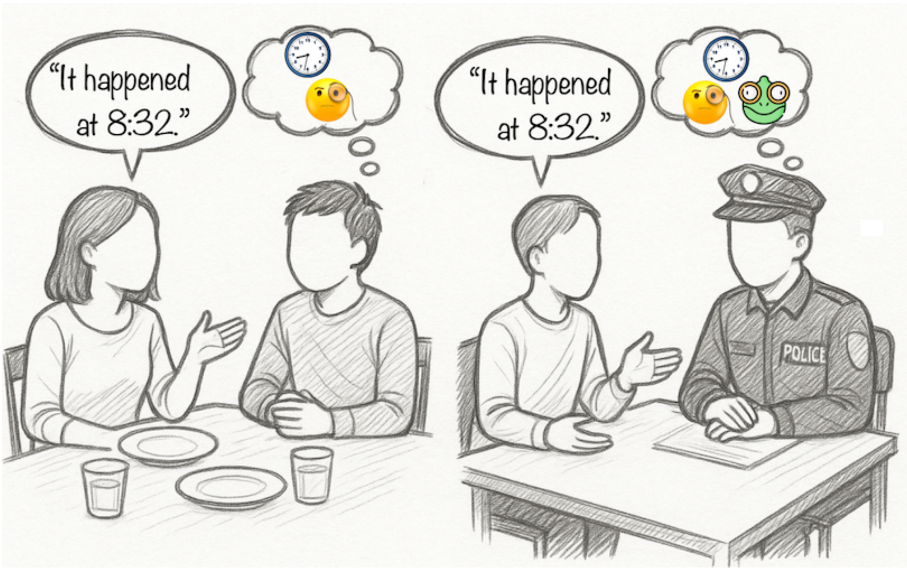
To fit the impact function I, we examine the Task 2 results. The full fitting procedure is outlined in Figure 7. First, we mapped the motivation classes from Task 2 (Figure 7(a)) to model parameters. For knowledge states: KNEWEXACT and INFOAVAIL

knowledge states $k \in K$	respective truthful utterances $u \in t(k)$
k_p	u_{apx} "It happened around 8:30." u_{prc} "It happened at 8:32."
$k_{\neg p}$	u_{apx} "It happened around 8:30."

(a) speaker's knowledge states and respective truthful utterances

	c_{HP} (police) k_p	c_{HP} (police) $k_{\neg p}$	c_{LP} (dinner) k_p	c_{LP} (dinner) $k_{\neg p}$
m_{max}	u_{prc}	u_{apx}	u_{prc}	u_{apx}
m_{sit}	u_{prc}	u_{apx}	u_{apx}	u_{apx}
m_{min}	u_{apx}	u_{apx}	u_{apx}	u_{apx}

(b) motivation-based strategies $m : C \times K \to V$



(c) model predictions for which knowledge states and motivation-based strategies a listener considers as probable after a speaker utters u_{prc} (left: dinner context; right: police context)

Figure 6: Exemplary scenario: a speaker reports the time of an event (8:32) to a friend at a dinner party (low-precision context) or a police officer (high-precision context). (a) Speaker knowledge states and respective truthful utterances. (b) The three motivation-based strategies: m_{max} (always maximally precise), m_{sit} (situation-adapted), m_{min} (always minimally precise). (c) Listener inferences after hearing a precise utterance, by context.

map to k_p; notKnow and infoNotAvail map to $k_{\neg p}$. For motivation-based strategies: fussyPerson (inherently precise) maps to m_{max}; easySpeaker and usualApx (consistent approximation for non-contextual reasons) map to m_{min}; needed, realWorld, and purpose (contextually-sensitive motivations) map to m_{sit}. All mappings are listed in Figure 7(c).

The trivalent functions ϕ_K and ϕ_M are then derived from these comparisons (Figure 7(b)): for each motivation class and attribute we assign $+1$ for a significant positive difference, -1 for a significant negative difference, and 0 otherwise (threshold $p < .01$),

Class	Description
KnewExact	S knew/was confident about exact/ precise information
NotKnow	S did not know, not sure/certain, not remember, approx. knowledge
FussyPerson	S is anal/predantic/detail oriented
EasySpeaker	using *apx* was easier/faster for S
UsualApx	S usually uses *apx*
Needed	utterance is needed, required, called for, important, necessary, or sufficient
RealWorld	real-world consequences of communicated information
Purpose	S is purposefully precise/approximate
InfoNotAvail	precise information was not available
InfoAvail	precise information was available

(a) The 10 most frequent motivation classes

	LIKEABLE	COMPETENT	PEDANTIC
KnewExact	.480	**<.001**	.028
NotKnow	.189	**<.001**	**<.001**
FussyPerson	.016	.033	**<.001**
EasySpeaker	.062	**.003**	.199
UsualApx	.933	.129	.106
Needed	.331	**.010**	.131
RealWorld	.365	.098	.857
Purpose	**<.001**	.034	.252
InfoNotAvail	.507	.851	.096
InfoAvail	.185	.116	.015

(b) p values for effects of motivation on feature evaluations

Parameter	Motivation class(es)
k_p	KnewExact, InfoAvail
$k_{\neg p}$	NotKnow, InfoNotAvail
m_{max}	FussyPerson
m_{min}	EasySpeaker, UsualApx
m_{sit}	Needed, RealWorld, Purpose

(c) Mappings from motivations to model parameters

	a_{like}	a_{comp}	a_{ped}
m_{max}	0	0	1
m_{sit}	1	1	0
m_{min}	0	-1	0

Fitted values for $\phi_M(m, a)$

	a_{like}	a_{comp}	a_{ped}
k_p	0	1	0
$k_{\neg p}$	0	-1	-1

Fitted values for $\phi_K(k, a)$

(d) Resulting trivalent functions

Figure 7: (a) The 10 most frequent motivation classes with descriptions. (b) p-values for associations between motivation classes and attribute ratings (COMPETENT, LIKEABLE, PEDANTIC); significant effects ($p < .01$) in bold, green (positive) or red (negative). (c) Mappings from motivation classes to model parameters. (d) Resulting trivalent functions ϕ_K and ϕ_M.

and replace motivation class labels with the corresponding model parameters per the mappings in Figure 7(c). Where several motivation classes map to the same parameter, the cell is set by majority vote. The resulting values are shown in Figure 7(d).[7]

3.4 Model Predictions

The fitted SEM has the following free parameters: weight $\omega \in (0, 1)$ in the impact function (Eq. 4), and the prior probabilities $P(k)$ and $P(m)$ used to compute the posteriors in Equations 2 and 3.

For an initial model check, we test a *balanced model*: $\omega = 0.5$ (equal weight to epistemic and motivational inference) and flat prior probabilities.[8] We then evaluate the model

[7]one cell, $\phi_M(m_{sit})$ on COMPETENT, lies at the significance boundary: the underlying NEEDED–COMPETENT difference is significant under a two-sample t-test ($p = .010$) but falls just short under a mixed-effects model with random intercepts for scenario ($p = .012$). We retain the t-test value; the model's four qualitative predictions hold under either choice.

[8]Specifically, $P(m_{max}) = P(m_{sit}) = P(m_{min}) = \frac{1}{3}$ and $P(k_p) = P(k_{\neg p}) = \frac{1}{2}$.

across a wide range of parameter combinations to assess robustness.

The model check tests whether SEM reproduces the four significant effects from Experiment 1 (all $p < .001$):

1. PRECISE is rated higher than APPROX on COMPETENT across all contexts.
2. PRECISE is rated higher than APPROX on PEDANTIC across all contexts.
3. The positive difference between PRECISE and APPROX on COMPETENT is greater in HPR than in LPR.
4. The relative strength of APPROX versus PRECISE on LIKEABLE is greater in LPR than in HPR.

SEM is considered qualitatively correct if it reproduces the *direction* of these effects. To state this precisely, and to introduce two measures we reuse for the quantitative evaluation in Section 4, we summarize each effect by a signed contrast on the model's evaluation scores. For the two main effects (1–2) we define, for an attribute a, the across-context form contrast

$$\Delta_a^{\text{main}} = \big[E_L(a|u_{prc}, c_{HP}) + E_L(a|u_{prc}, c_{LP})\big] - \big[E_L(a|u_{apx}, c_{HP}) + E_L(a|u_{apx}, c_{LP})\big],$$

and for the two interactions (3–4) the context-modulation contrast

$$\Delta_a^{\text{int}} = \big[E_L(a|u_{prc}, c_{HP}) - E_L(a|u_{apx}, c_{HP})\big] - \big[E_L(a|u_{prc}, c_{LP}) - E_L(a|u_{apx}, c_{LP})\big].$$

Each effect specifies a predicted sign for its contrast: positive for $\Delta_{comp}^{\text{main}}$ (Effect 1), $\Delta_{ped}^{\text{main}}$ (Effect 2), and $\Delta_{comp}^{\text{int}}$ (Effect 3), and negative for $\Delta_{like}^{\text{int}}$ (Effect 4).[9] We capture directional agreement with two scores, following Mühlenbernd (2026):

$$\text{DAS} = \frac{\#\{\text{significant main effects with correctly signed } \Delta^{\text{main}}\}}{\#\{\text{significant main effects}\}},$$

[9] Effect 4 is stated as the APPROX-over-PRECISE advantage on LIKEABLE being greater in LPR than HPR; written as $\Delta_{like}^{\text{int}}$ above, this is a negative contrast.

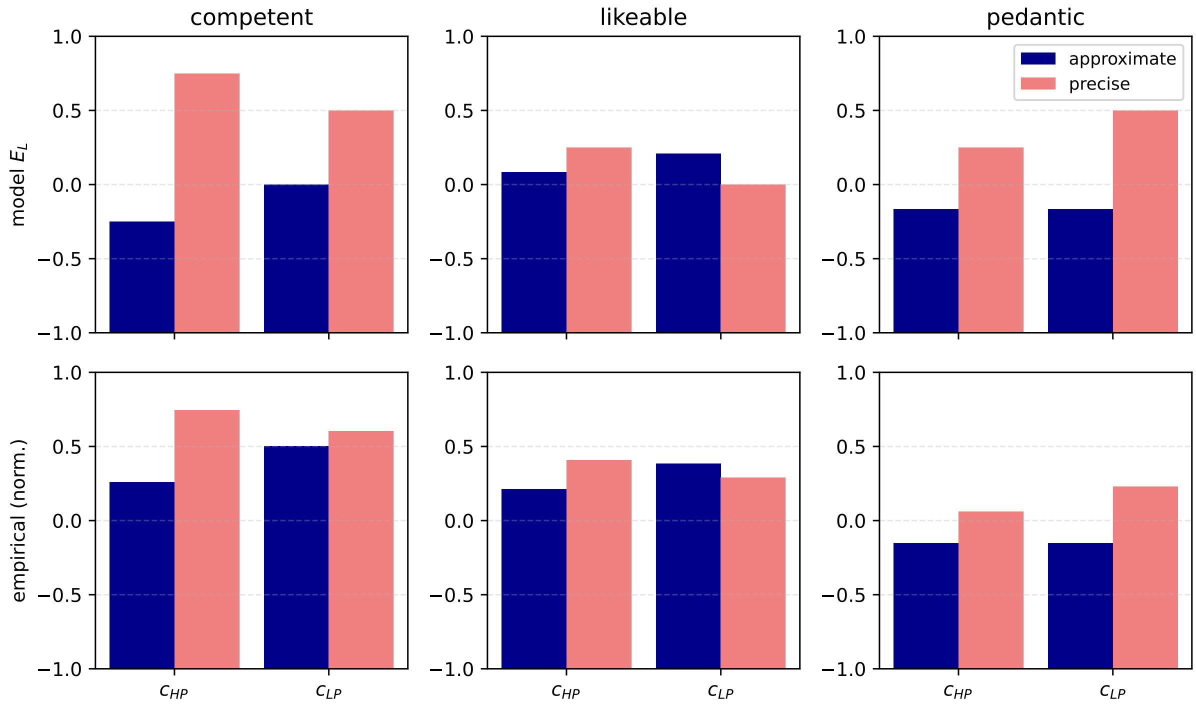


Figure 8: Predictions of the balanced model for the attributes COMPETENT, LIKEABLE, and PEDANTIC (top row, evaluation scores $E_L \in [-1, 1]$), shown alongside the empirical results of Experiment 1 (bottom row, mean ratings linearly rescaled to $[-1, 1]$ via $(x - 4)/3$). Bars rise from the scale floor (-1, the lowest evaluation); 0 denotes a median, not neutral, evaluation.

$$\text{ISS} = \frac{\#\{\text{significant interactions with correctly signed } \Delta^{\text{int}}\}}{\#\{\text{significant interactions}\}}.$$

The Directional Agreement Score (DAS) and Interaction Sensitivity Score (ISS) each range over $[0, 1]$, with 1 indicating that the model reproduces the sign of every significant effect of the relevant type. SEM is qualitatively correct on this case study exactly when DAS = ISS = 1. Figure 8 presents the balanced model's predictions for a_{comp}, a_{like}, and a_{ped} across utterances and contexts, alongside the empirical results. The balanced model achieves DAS = ISS = 1, reproducing the sign of all four significant effects, and the attribute-internal patterns are visibly aligned with the empirical data.

To assess robustness, we tested the balanced model against an exhaustive grid of free parameter combinations: every combination of ω (from 0.05 to 0.95) and the prior probabilities $P(k_p)$, $P(k_{\neg p})$, $P(m_{max})$, $P(m_{sit})$, and $P(m_{min})$ (from 0 to 1), each in increments of 0.02, yielding 2,593,080 valid combinations. SEM achieves DAS = ISS = 1 for 99.87% of combinations; for the remaining 0.13% one of the four effects flips sign (always within the main effects, so ISS = 1 throughout).[10] This demonstrates the robustness of the qualitative predictions across the free parameter space.

Section 4 presents a quantitative evaluation of these predictions, comparing the full model against nested ablations to assess which inferential components are necessary to reproduce the empirical patterns.

4 Quantitative Evaluation and Model Comparison

The preceding sections established that SEM qualitatively reproduces the four significant empirical effects from Experiment 1 across almost all free parameter settings (99.87% of 2,593,080 combinations). This section asks a deeper question: *how well* does the model reproduce those effects, and does its formal inference machinery add explanatory value over simpler alternatives?

We address this with two analyses. First, we evaluate the balanced model's quantitative fit against the empirical means using an established metric suite. Second, we compare the full model against three nested ablations, each of which removes one inferential component, evaluated across a sample of parameter combinations rather than a single setting, so that differences reflect structural properties of the model variants rather than parameter choices.

4.1 Metric Framework

Quantitative model evaluation follows the framework proposed by Mühlenbernd (2026) for assessing probabilistic pragmatic models against human data. We compare model scores $E_L(a|u,c) \in [-1,1]$ against empirical human means, which are linearly rescaled from the 7-point Likert scale to $[-1,1]$ via $(x-4)/3$, making model and human values directly comparable.

We report five metrics. The first two are the directional scores already introduced

[10] In particular, the model struggled to predict Effect 1 correctly when ω was very low, $P(k_p)$ was very low, and $P(m_{sit})$ was high, a combination of parameters that is difficult to motivate on independent grounds.

in Section 3; the remaining three quantify magnitude calibration and overall fit. For an attribute a, let $\Delta_{H,a}$ and $\Delta_{M,a}$ denote the human and model contrast values (the Δ_a^{main} or Δ_a^{int} of Section 3, as appropriate).

- **DAS** (Directional Agreement Score) and **ISS** (Interaction Sensitivity Score): as defined in Section 3: the proportion of significant main effects, respectively significant form $\times$ context interactions, on which model and human agree in sign. Range $[0, 1]$.
- **ESR** (Effect Size Ratio): for each *significant* effect, $|\Delta_{M,a}|/|\Delta_{H,a}|$. ESR $= 1$ means perfect magnitude calibration; ESR > 1 means the model overshoots. Computed only for significant effects to avoid division by near-zero human contrasts.
- **CDS** (Calibration Deviation Score): mean $|\text{ESR} - 1|$ across significant effects. Lower is better; $0 =$ perfect magnitude calibration.
- **RMSE**: root mean squared error computed cell-by-cell over all 12 condition means (3 attributes $\times$ 2 forms $\times$ 2 contexts), after rescaling human means to $[-1, 1]$. Lower is better.

The directional and calibration metrics (DAS, ISS, ESR, CDS) are computed over the four effects that are statistically significant in the human data: the main effects of form on a_{comp} and a_{ped}, and the form $\times$ context interactions on a_{comp} and a_{like}. The remaining two of the six possible effects, the main effect on a_{like} and the interaction on a_{ped}, are non-significant and are not assessed, since testing directional agreement or magnitude calibration on a near-zero human contrast would treat noise as signal. RMSE is the one exception: it is computed cell-by-cell over all twelve condition means rather than over effects, giving a global picture of fit across the full data surface.

4.2 Balanced Model Fit

Table 1 reports the full metric set for the balanced model ($\omega = 0.5$, flat priors). The focus here is on the calibration metrics ESR, CDS, and RMSE; DAS and ISS are included in Table 1 for completeness but, as both equal 1.00, were already discussed in Section 3.

Table 1: Quantitative fit of the balanced model ($\omega = 0.5$, flat priors) against empirical means from Experiment 1 ($N = 362$). DAS, ISS, ESR, and CDS are computed over the four statistically significant effects only; RMSE over all 12 conditions.

Metric	Value
RMSE	0.26
DAS	1.00
ISS	1.00
ESR main a_{comp} / a_{ped}	2.55 / 1.83
ESR interaction a_{comp} / a_{like}	1.32 / 1.31
CDS (main)	1.19
CDS (interaction)	0.31
CDS (overall)	0.75

The calibration metrics reveal a systematic pattern the directional check cannot see. The balanced model overshoots effect magnitudes on the main effects: ESR values of roughly 2× indicate that SEM predicts about twice the human contrast for both a_{comp} and a_{ped}, driving CDS (main) to 1.19. Calibration on the interactions is markedly better (CDS = 0.31, ESR $\approx$ 1.3 for both a_{comp} and a_{like}); that is, the model is best calibrated precisely where its context-sensitive reasoning does the most work, and overshoots on the simpler main effects.

The magnitude overshoot on main effects is a property of the balanced parameter setting rather than a structural limitation of the model. A grid search over the free parameter space identifies a setting ($\omega = 0.7$, $P(k_p) = 0.9$, $P(m_{max}) = 0.5$, $P(m_{sit}) = 0.3$, $P(m_{min}) = 0.2$) that achieves near-perfect calibration (CDS = 0.056, ESR values of 1.03, 1.01, 1.15, 0.96; RMSE = 0.165), while preserving DAS = ISS = 1.00. This confirms that the overshoot under balanced parameters reflects the uninformative prior choice, not an architectural flaw.[11]

4.3 Model Variant Comparison

A central question for any model of this kind is whether its inference machinery adds explanatory value beyond simpler alternatives. We address this by asking which of SEM's

[11] This is an in-sample, post-hoc result: the optimized parameters are tuned to the same 12 data cells from which the impact functions were derived, so it represents an upper bound on fit rather than a validated generalization. The balanced model is the primary evaluation throughout; the grid search result is reported as a calibration bound.

inferential components are necessary to reproduce the empirical pattern, comparing the full model against three nested ablations, each of which removes one inferential component while holding everything else equal:

- **SEM-full**: the complete model, all parameters free.
- **SEM-context-blind**: $P(m_{sit}) = 0$, removing the context-sensitive motivation strategy. This operationalizes a "violation-of-expectation" account in which the model can infer knowledge states and non-contextual motivations, but has no mechanism for context to modulate evaluations positively.
- **SEM-knowledge-only**: $\omega = 1$, so only epistemic inference contributes to evaluation. Motivation inference is disabled entirely.
- **SEM-motivation-only**: $\omega = 0$, so only motivational inference contributes. Knowledge-state inference is disabled entirely.

To make the comparison robust to parameter choice, each variant is evaluated across 100 sampled parameter combinations rather than a single setting. Free parameters are drawn from uninformative distributions ($\omega \sim \text{Beta}(2, 2)$ clipped to $(0.05, 0.95)$; $P(k) \sim \text{Dirichlet}(1, 1)$; $P(m) \sim \text{Dirichlet}(1, 1, 1)$; seed fixed for reproducibility), and metrics are computed for each draw. We report mean $\pm$ standard deviation across draws. A difference that is consistent across the full sample reflects a structural property of the model variant, not a parameter artifact.

Table 2 reports the results.

The key finding is ISS: it collapses to ≈ 0 for the two variants that remove m_{sit} (context-blind: 0.00; knowledge-only: 0.04) and stays at 1.00 for the two that retain it (full and motivation-only). Since the two significant interaction effects are precisely the form $\times$ context effects captured by m_{sit}, this means context-blind and knowledge-only cannot reproduce two of the four significant empirical effects at all, regardless of parameter setting. The context-sensitive motivation mechanism is structurally necessary, not a decorative feature of the model.

Table 2: Quantitative comparison of four SEM variants across 100 sampled parameter combinations (mean ± SD). DAS, ISS, ESR, and CDS are computed over the four significant effects only; RMSE over all 12 conditions. Lower CDS and RMSE are better; higher ISS and DAS are better.

Metric	SEM-full	Context-blind	Know-only	Motiv-only
DAS	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.96 ± 0.13
ISS	$\mathbf{1.00 \pm 0.00}$	0.00 ± 0.00	0.04 ± 0.14	1.00 ± 0.00
RMSE	$\mathbf{0.30 \pm 0.10}$	0.42 ± 0.12	0.39 ± 0.14	0.51 ± 0.09
CDS (main)	$\mathbf{1.06 \pm 0.71}$	1.46 ± 0.72	1.65 ± 1.25	1.07 ± 0.56
CDS (inter)	$\mathbf{0.82 \pm 0.62}$	1.00 ± 0.00	1.00 ± 0.00	1.79 ± 1.44
CDS (overall)	$\mathbf{0.94 \pm 0.43}$	1.23 ± 0.36	1.32 ± 0.62	1.43 ± 0.67

Each ablation fails on a distinct dimension, and none matches the full model on both structure and calibration. The context-blind variant loses the interactions entirely (ISS = 0) and also degrades on RMSE ($0.30 \rightarrow 0.42$) and overall CDS ($0.94 \rightarrow 1.23$). The motivation-only variant keeps the interaction directions (ISS = 1.00) but is otherwise the weakest: it massively overshoots interaction magnitudes (mean ESR ≈ 2.7, CDS interaction = 1.79) and has the highest RMSE (0.51), because without knowledge inference the competence ordering breaks down. The knowledge-only variant achieves the second-lowest RMSE (0.39), reflecting that the study's overall ordering is largely epistemic: precise speakers are inferred to have precise knowledge, which is the dominant signal in the data. But its ISS of 0.04 means it fails entirely on the interactions. Without motivational inference, the model cannot represent context as modulating the epistemic ordering, and so it misses two of the four significant empirical effects regardless of parameter setting. A model that reproduces the main trend while missing the context-sensitivity has not captured the phenomenon.

Taken together, the full SEM is the only variant that reproduces the interaction structure (ISS = 1.00) while remaining competitive on global fit (lowest RMSE, lowest overall CDS). Every simpler alternative either matches or exceeds it on one metric while failing conspicuously on another. This pattern provides direct evidence that the model's staged inference, from epistemic state through contextual motivation, to social attribute, is necessary rather than redundant.

The CDS results also bear on the question of magnitude calibration raised above. The interaction CDS difference between the full model (0.82) and the ablations that lose m_{sit} (both pinned at 1.00, since their predicted interaction contrasts are zero) quantifies the specific contribution of contextual motivation inference to calibration. The full model's main-effect CDS (1.06) is essentially tied with the motivation-only variant (1.07), confirming that main-effect overshoot is a shared property of the trivalent impact scale and not specific to any particular model configuration.

The symmetry of the two inferential dimensions. It would be a misreading of these results to conclude that one inferential component is more fundamental than the other. The ablations show that for *this* phenomenon both the epistemic and the motivational dimension are required, but they do not rank the two: knowledge-only fails on the interactions (ISS) and motivation-only fails on calibration, because numerical (im)precision happens to load on both dimensions at once, the epistemic dimension securing the competence ordering and the motivational dimension securing its context-sensitivity.

Other phenomena need not be balanced in this way. SEM can represent a phenomenon in which either dimension is inert, and it can do so in two ways that require no change to the model definition. One is through the weight: setting $\omega = 1$ or $\omega = 0$ removes the motivational or the epistemic contribution to the impact function, respectively, which is exactly what the knowledge-only and motivation-only ablations do. The other, and more natural where a distinction is simply irrelevant, is through the inference domains themselves: a phenomenon in which the speaker's knowledge is not at issue is modeled with a single knowledge state ($|K| = 1$), and one in which no motivational distinction is relevant with a single strategy ($|M| = 1$). In either case the corresponding inference becomes vacuous and the social evaluation is driven entirely by the other dimension. A purely epistemic phenomenon, such as the social uptake of a scalar implicature, where what matters is whether the speaker is taken to know the stronger alternative, would be a natural $|M| = 1$ case; a phenomenon in which the speaker's knowledge is not in question but their communicative motivation is, as in some politeness and register choices (Section 5.4), would be a natural $|K| = 1$ case. The full SEM is required for (im)precision not because the architecture privileges any one route, but because this particular phenomenon exercises both.

Threshold robustness. The ϕ functions are derived at $p < .01$. Re-running the full comparison with ϕ derived at $p < .05$ (which adds three non-zero cells) worsens the full model on every calibration metric (CDS 0.94 $\rightarrow$ 1.45, RMSE 0.30 $\rightarrow$ 0.51) while leaving ISS unchanged. Tightening to $p < .001$ slightly improves CDS (0.94 $\rightarrow$ 0.89) but degrades interaction coverage (ISS 1.00 $\rightarrow$ 0.57) because the borderline m_{sit}–a_{comp} cell is dropped. The $p < .01$ threshold is the only one that maintains full interaction coverage while achieving competitive calibration.[12]

5 Discussion and Conclusion

5.1 Summary of Contributions

The Social Evaluation Model (SEM) is a probabilistic framework that explains how listeners form social impressions of speakers based on their pragmatic behavior, integrating cognitive, epistemic, and social reasoning within a unified formal architecture. By extending the inferential chain of probabilistic pragmatics, SEM shows how linguistic choices, such as the level of numerical precision, give rise to evaluations of traits like competence, warmth, and pedantry through staged Bayesian inference. The model accurately reproduces the key empirical patterns from a matched-guise study on numerical (im)precision: it predicts how varying levels of precision affect competence, likeability, and pedantry ratings as a function of contextual demands, and the quantitative evaluation demonstrates that its staged inference architecture, from epistemic state through contextual motivation, to social attribute, is necessary to reproduce the observed pattern. Three nested ablations each fail on a structurally distinct dimension, with the context-blind variant providing the clearest evidence: removing the context-sensitive motivation strategy collapses interaction sensitivity to zero, directly demonstrating that the m_{sit} mechanism is not dec-

[12] These threshold comparisons were conducted by re-deriving ϕ_K and ϕ_M at each threshold from the same Experiment 1 data and re-running the variant comparison. Scripts are available in the repository.

orative. More broadly, the model's staged inference architecture is transparent, testable, and quantitatively evaluable, and the variant comparison provides direct evidence that each inferential component contributes independently to the model's predictions.

5.2 Relation to the RSA Framework

The relationship between SEM and the Rational Speech Act framework (Frank and Goodman, 2012; Goodman and Frank, 2016) deserves careful discussion. A recurring question is whether a new model is needed, or whether the existing RSA architecture could simply be extended to cover social evaluation. Our position is that SEM is best understood as a distinct model that *extends* the RSA inferential chain, it inherits RSA's core machinery of Bayesian listener inference over utterance alternatives, but adds the components needed to make social evaluation an explicit target of inference, and that it is naturally compatible with, rather than opposed to, RSA.

RSA is a framework for modeling how communicative meaning is derived through recursive reasoning about speaker and listener mental states. In standard RSA, the speaker selects utterances to maximize a utility function, typically a function of informational value and cost; the listener uses Bayes' rule to invert this speaker model and recover the intended meaning. Social goals can be incorporated into the utility function, as in Yoon et al.'s (2020) model of polite speech, which treats politeness as a competing goal alongside informativeness. This shows that RSA can be extended to encompass some aspects of social reasoning, and we take this as an important point of contact.

SEM differs, however, in its architecture and target of inference. First, SEM models social evaluation from the *listener's* perspective, whereas RSA typically models the speaker's production choices. In SEM, the listener's inferential process yields a social evaluation as its output; the model does not make claims about how speakers optimize their utterances (though see below for how SEM and RSA can be integrated). Second, SEM replaces the continuous utility function of RSA with a discrete set of behavioral strategies, each representing a recognizable communicative motivation. This design choice makes the strategies directly groundable in empirical data, specifically in the motivation classes that participants attribute to speakers in free-text tasks, and makes the model's parameters interpretable and testable. A utility-based extension of RSA could also include social attributes as latent variables, but the mapping from utilities to the behavioral diversity captured by SEM's motivation strategies would need to be explicitly specified. Third, the impact function I and the fitted trivalent functions ϕ_K and ϕ_M have no counterpart in standard RSA. They constitute the formal machinery for linking pragmatic inference to social evaluation, and their empirical grounding is a methodological contribution distinct from RSA extensions.

Taken together, SEM is best understood not as an alternative to RSA but as a complementary module that extends the inferential chain one step further: from what the speaker meant, to what kind of communicator they are.

5.3 Relation to the Social Meaning Game

SEM also relates to the Social Meaning Game (SMG) framework (Burnett, 2017; Burnett, 2019; Burnett, 2023). SMG models how linguistic variation acquires indexical social meaning through game-theoretic reasoning: speakers and listeners reason about the mapping between linguistic forms and social categories, and social meaning arises from learned indexical links between form and identity.

SEM and SMG are complementary in scope. SMG is designed for semantically equivalent variants whose social meaning is form-based (e.g., sociophonetic variation such as ING); SEM is designed for semantically non-equivalent alternatives where social meaning arises from pragmatic inference about the speaker's epistemic state and motivation. SMG explains how forms come to index social categories; SEM models how differences in semantic content or pragmatic choice lead listeners to attribute specific character traits. Correspondingly, SMG is better suited to explaining the acquisition and conventionalization of social meanings over time, while SEM is better suited to explaining how novel or context-sensitive pragmatic choices give rise to social evaluations in the moment of interpretation.

5.4 Extending SEM to Other Social-Meaning Phenomena

Although we have developed and tested SEM on a single phenomenon, numerical (im)precision, its architecture is not specific to that domain. SEM applies, in principle, wherever three conditions hold: a speaker chooses among non-equivalent alternatives, that choice supports inferences about the speaker's knowledge state and communicative motivation, and those inferences in turn bear on the listener's social evaluation of the speaker. A number of well-studied phenomena meet these conditions, and we sketch several here as candidates for future modeling.

The most direct extension is to the pragmatic violations studied by Beltrama and Papafragou (2023), who show that a speaker's adherence to or violation of the Gricean maxims of Relevance and Informativeness shifts listeners' evaluations of the speaker's warmth and competence, and, crucially, that the *reason* attributed to a violation modulates this effect: an uninformative answer attributed to inability is evaluated differently from one attributed to unwillingness. This is precisely the structure SEM formalizes. The alternatives are the relevant versus irrelevant, informative versus uninformative responses; the knowledge states distinguish a speaker who cannot be informative from one who can; and the motivation strategies capture why a speaker would withhold relevant information. A SEM treatment would represent inability and unwillingness as distinct motivation-based strategies with different impact-function signatures on warmth and competence, yielding the asymmetry Beltrama and Papafragou observe. Because their study elicits social-attribute ratings on the same warmth and competence dimensions used here, it is well suited to the quantitative evaluation of Section 4.

Politeness and indirect speech offer a second candidate. A speaker who chooses an indirect or mitigated form over a direct one invites inferences about their motives, deference, face-management, or self-presentation, that feed social evaluation (Brown and Levinson, 1987; Lee and Pinker, 2010). Probabilistic models of polite language already treat such choices as arising from competing informational and social goals: in Yoon et al.'s (2020) model, a speaker selects utterances partly to manage how they will be perceived, treating *self-presentation* as an explicit production goal. This is revealing for SEM, because a speaker's self-presentation goal just is an anticipation of the social evaluation a listener will form: the speaker reasons about the very inference that SEM models on the listener's side. The two models thus describe complementary halves of a single social-pragmatic loop, the speaker choosing a form in anticipation of being evaluated, the listener evaluating on the basis of that choice, and SEM supplies the listener-side component that a self-presenting speaker must implicitly represent. We return to this speaker–listener integration in Section 5.6.

A third candidate is the use of *hedging* and uncertainty expressions (e.g., *I think*, *probably*, *might*). The choice between a hedged and an unhedged assertion is a choice among non-equivalent alternatives that supports inferences about the speaker's epistemic state and their motivation for marking it, and these inferences carry social weight. McCready (2015) analyzes hedging precisely as a reputation-management device: speakers hedge to protect their standing for cooperativity against the cost of an assertion that may turn out to be unwarranted. Cast in SEM's terms, this is a motivation strategy whose impact-function signature trades perceived competence (a confident, unhedged assertion projects knowledge) against the social protection a hedge affords if the speaker is wrong, with the social outcome depending on which motivation the listener infers. Listeners are demonstrably sensitive to such variation and adapt to speaker-specific patterns of uncertainty marking (Schuster and Degen, 2020), which also makes hedging a natural site for studying the listener variation that SEM's free parameters are designed to represent.

A fourth candidate, drawn from a different grammatical domain, is honorification and register. McCready (2019) analyzes honorifics as encoding the formality a speaker takes to be appropriate to the speech situation; the choice among register levels is thus a context-sensitive selection among alternatives that conveys social meaning. Where an honorific form is over- or under-used relative to situational expectations, listeners draw inferences about the speaker's stance, deference, or social competence, an inferential step of exactly the kind SEM models, though here the relevant attributes (e.g., respectfulness, social fluency) and the role of situational context differ from the precision case.

What unites these phenomena is the structure SEM is built to capture: a choice among meaningful alternatives, interpreted through inferences about what the speaker knows and why they chose as they did, issuing in a social evaluation. Establishing SEM's generality across them, ideally with the same quantitative methodology applied here, is a central goal for future work.

5.5 Limitations

Several limitations of the current formulation are worth noting. The trivalent functions ϕ_K and ϕ_M make a simplifying assumption: that a given knowledge state or motivation-based strategy has a fixed, consistent impact on a social attribute regardless of context. In reality, the underlying motivations corresponding to a single behavioral strategy may differ in their social implications. A more refined model would include an additional inference step recovering the probability of each specific motivation given a strategy; we leave this generalization to future work. The current model has been evaluated on a single empirical case study (numerical imprecision); the framework's applicability to other pragmatic domains, for example politeness phenomena or relevance violations, is a natural direction for future empirical work.

A further simplification concerns the set of social attributes A itself. SEM takes A to be a finite set supplied by the modeler, here the attributes elicited in the experiment, but in principle the attributes a listener might ascribe to a speaker are open-ended. The model is agnostic about which attributes populate A: any attribute for which the impact functions ϕ_K and ϕ_M can be specified (whether fitted from data or stipulated) can be added without change to the architecture. What SEM does not currently model is how listeners select or generate the relevant attributes in the first place; we take the attribute set as given and model the inference from utterance to evaluation along it.

A limitation of the current formulation concerns the structure of listener variation. As noted in Section 2, the priors and weight ω already encode listener-level variation, and the parameter sampling in Section 4.3 treats a spread of settings as a population of listeners. What this does not capture is *systematic* variation: for instance, speaker-specific priors arising from the sensitivity of comprehension to speaker identity, or differences in ω tied to listener traits or social relationships. Such structure could be incorporated by conditioning the priors or the weight on listener- and speaker-level covariates, and modeling how these are themselves inferred; we leave this to future work.

5.6 Integration with RSA and Future Directions

Looking forward, SEM can be naturally integrated into higher-order models of pragmatic reasoning such as RSA. In such an integrated model, the outcomes of SEM, listeners' judgments of competence, warmth, or sincerity, become part of the speaker's production process: speakers could anticipate the evaluative inferences their utterances are likely to elicit, and strategically choose expressions that present them favorably. This connects pragmatic reasoning to social signaling and impression management, framing communication as recursive social cognition in which agents reason continuously about both the informational and social consequences of their linguistic choices. Developing this integration formally, and testing its predictions empirically, is a central direction for future research.

Data and code availability. The experimental data and R analysis scripts are available on the Open Science Framework at `[ANONYMIZEDOSFPROJECTLINK]`; the preregistration for Experiment 1 is at `[ANONYMIZEDOSFPREREGLINK]`. A Python implementation of SEM, including a Jupyter notebook that reproduces all analyses and figures reported here, is available at `[ANONYMIZEDGITHUBREPOLINK]`.

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